

Supplementary Tables: Full ROI Benchmark Results for 46 Urban Settlement Indices

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Description: This supplementary file contains the full pixel level performance results for all 46 urban settlement indices evaluated over the Bahria Town Phase 7 region of interest. The benchmark uses Landsat-8 OLI and TIRS, VIIRS DNB nighttime lights, and a Dynamic World Built reference mask resampled to 30 m. The table reports accuracy, precision, recall, F1 score, IoU score, TP, TN, FP, FN, and the NDVI gating threshold used for each index.

Supplementary Table S16: Pixel level performance of urban indices over the Bahria Town Phase 7 ROI using Landsat-8 OLI and TIRS, VIIRS DNB, and a Dynamic World Built mask resampled to 30 m. Built up is the positive class. Each row reports accuracy, precision, recall, F1 score, and IoU score as percentages, together with TP, TN, FP, and FN pixel counts and the NDVI gating threshold used for that index.

No.	Index	Accuracy (%)	Precision (%)	Recall (%)	TP	TN	FP	FN	F1 (%)	IoU (%)	NDVI thr.
<i>Multispectral indices</i>											
1	NDBI	91.70	89.66	99.67	15018	4642	1731	49	94.40	89.40	0.70
2a	IBI with NDVI	91.49	89.29	99.87	15047	4568	1805	20	94.28	89.18	0.75
2b	IBI with SAVI	91.80	90.01	99.36	14970	4712	1661	97	94.45	89.49	0.75
3	BAEI	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
4	BBI	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
5	BCI	91.91	90.09	99.41	14979	4726	1647	88	94.52	89.62	0.70
6	BLFEI	92.19	90.46	99.36	14971	4794	1579	96	94.70	89.94	0.70
7	BU_b	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
8	BU_c	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
9	BUI	92.14	90.51	99.22	14950	4806	1567	117	94.67	89.88	0.70
10	CBCI	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
11	CBI	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
12	ENDISI	91.76	90.58	98.51	14843	4830	1543	224	94.38	89.36	0.65
13	MBAI	92.95	91.34	99.39	14975	4953	1420	92	95.19	90.83	0.75
14	NBAI	91.55	89.40	99.80	15037	4591	1782	30	94.32	89.24	0.75
15	BRBA	92.15	90.52	99.22	14949	4808	1565	118	94.67	89.88	0.70
16	NBI	91.58	89.69	99.46	14986	4650	1723	81	94.32	89.25	0.70
17	PISI	91.55	89.42	99.77	15033	4595	1778	34	94.32	89.24	0.70
18	RNDISI	91.50	89.30	99.87	15048	4570	1803	19	94.29	89.20	0.75
19	SwiRed	92.78	91.07	99.48	14989	4904	1469	78	95.09	90.64	0.75
20	UI	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
21	VIBI	91.70	89.59	99.77	15033	4627	1746	34	94.41	89.41	0.75
22	VrNIR-BI	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
23	VgNIR-BI	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
24	WE-NDBI	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
25	UBI	91.49	89.29	99.87	15048	4568	1805	19	94.28	89.19	0.75
<i>Panchromatic indices</i>											
26	MNDISI	92.75	91.07	99.43	14981	4905	1468	86	95.07	90.60	0.75
27	RUI	91.49	89.29	99.87	15047	4568	1805	20	94.28	89.18	0.75

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Supplementary Table S16: Pixel level performance of urban indices over the Bahria Town Phase 7 ROI, continued.

No.	Index	Accuracy (%)	Precision (%)	Recall (%)	TP	TN	FP	FN	F1 (%)	IoU (%)	NDVI thr.
28	NRUI	91.49	89.29	99.87	15047	4568	1805	20	94.28	89.18	0.75
<i>Thermal indices</i>											
29a	NDISI with VIS	91.49	89.29	99.87	15047	4568	1805	20	94.28	89.18	0.75
29b	NDISI with NDWI	92.05	90.03	99.74	15028	4709	1664	39	94.63	89.82	0.75
29c	NDISI with MNDWI	92.90	91.19	99.51	14994	4925	1448	73	95.17	90.79	0.75
30	EBBI	91.49	89.29	99.87	15047	4568	1805	20	94.28	89.18	0.70
31	STRED	92.80	91.24	99.29	14960	4936	1437	107	95.09	90.64	0.75
32	NDII	91.55	89.44	99.75	15030	4598	1775	37	94.31	89.24	0.75
33	DBI	91.49	89.29	99.87	15047	4568	1805	20	94.28	89.18	0.70
34	MNDISI	91.62	89.85	99.30	14962	4682	1691	105	94.34	89.28	0.75
35	NDBI OLI	91.48	89.29	99.86	15046	4568	1805	21	94.28	89.18	0.75
36	BAEM	91.63	89.52	99.77	15032	4614	1759	35	94.37	89.34	0.70
<i>Nighttime light indices</i>											
37	HSI	91.66	89.60	99.71	15023	4629	1744	44	94.38	89.36	0.75
38	VANUI	91.58	89.75	99.36	14971	4664	1709	96	94.31	89.24	0.75
39	LISI	91.60	89.70	99.47	14987	4653	1720	80	94.33	89.28	0.75
40	NDUI	91.91	90.15	99.35	14969	4737	1636	98	94.53	89.62	0.75
41	EUBI	91.53	89.60	99.48	14989	4634	1739	78	94.29	89.19	0.75
42	NUACI	91.53	91.53	96.92	14603	5021	1352	464	94.15	88.94	0.65
43	NAISI	92.14	91.25	98.23	14801	4954	1419	266	94.61	89.78	0.75
<i>Complex indices</i>											
44	MNDISI _{Liu}	91.48	89.29	99.86	15046	4568	1805	21	94.28	89.18	0.75
45	VTLI	91.67	89.96	99.22	14950	4705	1668	117	94.37	89.33	0.75
46	TVANUI	91.58	89.70	99.44	14982	4653	1720	85	94.32	89.25	0.75